

Neuropsychiatric symptoms in Parkinson's disease: review of data from the Arab world

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**Alfred Chabbouh¹, Hassan Khalil²,
Chafic Abboud¹, Antoine Sader¹  and
Rita Khoury¹ **

Abstract

Neuropsychiatric symptoms in Parkinson's disease (PD) are common, underrecognized, and impact patients' quality of life. Research around this topic is scarce in the Arab world. The aim of this comprehensive narrative review was to examine available evidence and discusses its strengths and limitations. The PubMed and Embase databases were searched using relevant keywords for studies published between 2000 and 2023. A total of 10 epidemiological and three assessment tool validation studies of neuropsychiatric symptoms in PD, conducted in the 22 Arab countries, were included. Only four Arab countries contributed to the 10 epidemiological studies found. Egypt provided the largest number of patients. Prevalence rates of neuropsychiatric symptoms in PD were heterogenous. Three studies relied on the Non-Motor Symptom Scale to estimate prevalence. Three studies used validated screening tools for dementia. Five studies used specific screening tools for depression, while two used the Diagnostic and Statistical Manual of mental disorders, fourth edition (DSM-IV) diagnostic criteria for depression. No appropriate and validated tool is available to assess for apathy. The generalizability of the findings is limited and recommendations for future studies are provided.

Keywords

Arab, Neuropsychiatric, Scales, Parkinson, Prevalence, Screening, Symptoms, Validation

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Corresponding author:

Rita Khoury, Department of Psychiatry, Faculty of Medicine & Medical Center, American University of Beirut, Beirut, Lebanon, PO Box 11-0236 Riad El Solh, Beirut 1107 2020 Beirut – Lebanon Location: Cairo Street, Hamra, Beirut, Lebanon.
Email: rk224@aub.edu.lb

¹Department of Psychiatry, Faculty of Medicine & The Medical Center, American University of Beirut, Beirut, Lebanon

²Faculty of Medicine, American University of Beirut, Beirut, Lebanon



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Introduction

Parkinson's disease (PD) is a neurodegenerative disease characterized by dopaminergic neuron loss in the substantia nigra. This loss leads to prominent motor symptoms including resting tremors, rigidity, bradykinesia, and gait abnormalities.¹ While age is the biggest risk factor for developing PD, prevalence varies worldwide, with higher rates in European and American populations.¹ The estimated prevalence is 23–43 per 100 000 individuals in the Arab world,¹ and 0.36% in Lebanon.² The incidence of PD has been increasing worldwide, with some authors suggesting this rise is partly a byproduct of the industrial revolution.³ The increasing incidence appears to be slower in Arab countries compared with the worldwide rate.⁴

While motor symptoms are key features of PD, non-motor symptoms (NMS), such as autonomic dysfunction, sexual dysfunction, hyposmia, fatigue, and neuropsychiatric symptoms, have been increasingly recognized.⁵ These symptoms are relatively common and can manifest at any stage of the disease, sometimes even before the onset of motor symptoms.⁶ For instance, depression, rapid-eye movement (REM) sleep behavior disorder, and excessive daytime sleepiness, are common premotor symptoms of PD that occur with other symptoms, such as olfactory impairment and constipation.¹

Among the NMS, neuropsychiatric manifestations of PD are often underrecognized by patients and clinicians alike and are undertreated in clinical practice.⁶ They include apathy, anxiety, depression, perceptual disturbances or more broadly psychotic symptoms, impulse control disorders (ICDs), sleep disorders, and cognitive impairment. The cumulative number and severity of neuropsychiatric symptoms in

patients with PD are associated with a substantial impact on patients' quality of life.⁷

The Arab world consists of 22 countries who are members of the League of Arab States,⁸ and comprises the geographical region present in Southwest Asia and North Africa (SWANA) that shares cultural and linguistic similarities.⁹ In 2022, the World Bank estimated an Arab world population of almost 465 million with an annual population growth of 1.8%. Among members of the League of Arab States, Qatar, United Arab Emirates (UAE), Bahrain, Kuwait, and Saudi Arabia are considered to be higher-income countries.¹⁰

As the life expectancy of patients with PD increases in the Arab world, researchers are anticipating a higher occurrence of neuropsychiatric symptoms, with further increase in disease burden and disability,⁴ while research and knowledge of PD in the Arab world remains scarce. Within a span of 15 years (2005–2019), only 253 articles related to PD were published by the 22 Arab countries combined. Research around the neuropsychiatric symptoms associated with PD is even more scant, despite their substantial prevalence and impact.¹¹ Therefore, the aim of the present study was to review the available literature in the Arab world concerning neuropsychiatric symptoms in patients with PD, in order to summarize data on the estimated prevalence rates, as well as the validated or unvalidated assessment tools used. The limitations of published articles are discussed and recommendations for future research investigations on neuropsychiatric symptoms in PD in the Arab world are proposed.

Materials and methods

The PubMed and Embase databases were searched for articles published between 2000 and 2023 from the 22 Arab countries, using the following keywords and structure:

(Country Name) AND 'PARKINSON' AND (neuropsychiatric symptom). Country names were 'Algeria', 'Arab', 'Bahrain', 'Comoros', 'Djibouti', 'Egypt', 'Iraq', 'Jordan', 'Kuwait', 'Lebanon', 'Libya', 'Mauritania', 'MENA', 'Morocco', 'Oman', 'Palestine', 'Qatar', 'Saudi Arabia', 'Somalia', 'Sudan', 'SWANA', 'Syria', 'Tunisia', 'United Arab Emirates', and 'Yemen'. Neuropsychiatric symptom keywords were: 'Anxiety', 'Apathy', 'Cognitive', 'Dementia', 'Depression', 'Impulse Control', 'Neuropsychiatric', 'Nonmotor', 'Perception', 'Psychosis', and 'Sleep'.

The rationale behind limiting the search to a 24-year period was that attention to neuropsychiatric symptoms in PD started in the 21st century. For example, the Non-Motor Symptoms Scale for PD – one of first tools developed – was initially published in 2007. Inclusion criteria were: original investigations that explored prevalence of neuropsychiatric symptoms in patients with PD, or studies validating Arabic versions of assessment tools of neuropsychiatric symptoms in the present population of interest.

The neuropsychiatric symptoms included in the search were depression, anxiety, apathy, psychosis, cognitive impairment, impulse control disorders, and sleep disorders. Researchers (AC and HK) initially and independently screened the titles and abstracts for relevant articles, followed by full text review (divided between all authors). Conflicts were resolved by one researcher (RK). Article reference lists were also screened for relevant studies that might be included in the review. Articles that met the inclusion criteria were included, while studies outside the scope of this review i.e., studies not validating any relevant tools or using tools to assess prevalence (e.g., genetic studies) were excluded. The article was produced according to the

scale for the assessment of narrative review articles (SANRA).¹²

Results

The screening process leading to the inclusion of 13 studies is summarized in Figure 1.

A total of three validation studies of Arabic instruments that assess for neuropsychiatric symptoms in PD were included, summarized in Table 1.^{13–15} The first was a multisite validation study performed in eight Arab countries (Jordan, Saudi Arabia, Tunisia, Egypt, Kuwait, Bahrain, Palestine, and UAE) validating the Movement Disorder Society Unified Parkinson's Disease Rating Scale (MDS-UPDRS) in 441 Arab-speaking patients with PD. The MDS-UPDRS evaluates all domains of PD (motor and non-motor) including neuropsychiatric symptoms.¹³ The second study validated the Arabic version of the Non-Motor Symptom Scale (NMSS) in 62 patients in Razi Hospital in Tunisia,¹⁴ which evaluates the following domains: sleep/fatigue, mood/cognition, perceptual disturbances/hallucinations, attention/memory, and autonomic dysfunction, among others.¹⁶ The third study validated the Parkinson's Disease Sleep Scale Revised Version (PDSS-2) in 133 patients with PD attending two neurology outpatient clinics in Egypt.¹⁵ No other validation studies were found regarding specific instruments targeting individual neuropsychiatric symptoms in Arab patients with PD.

A total of 10 original investigations that reported on the epidemiology/prevalence of neuropsychiatric symptoms in Arab patients diagnosed with PD were included. The studies were conducted in Egypt (six studies), Morocco (two studies), UAE (one study) and Lebanon (one study), and are summarized in Table 2.^{7,17–25} One Emirati study, while epidemiological in nature, did not report on prevalence rates

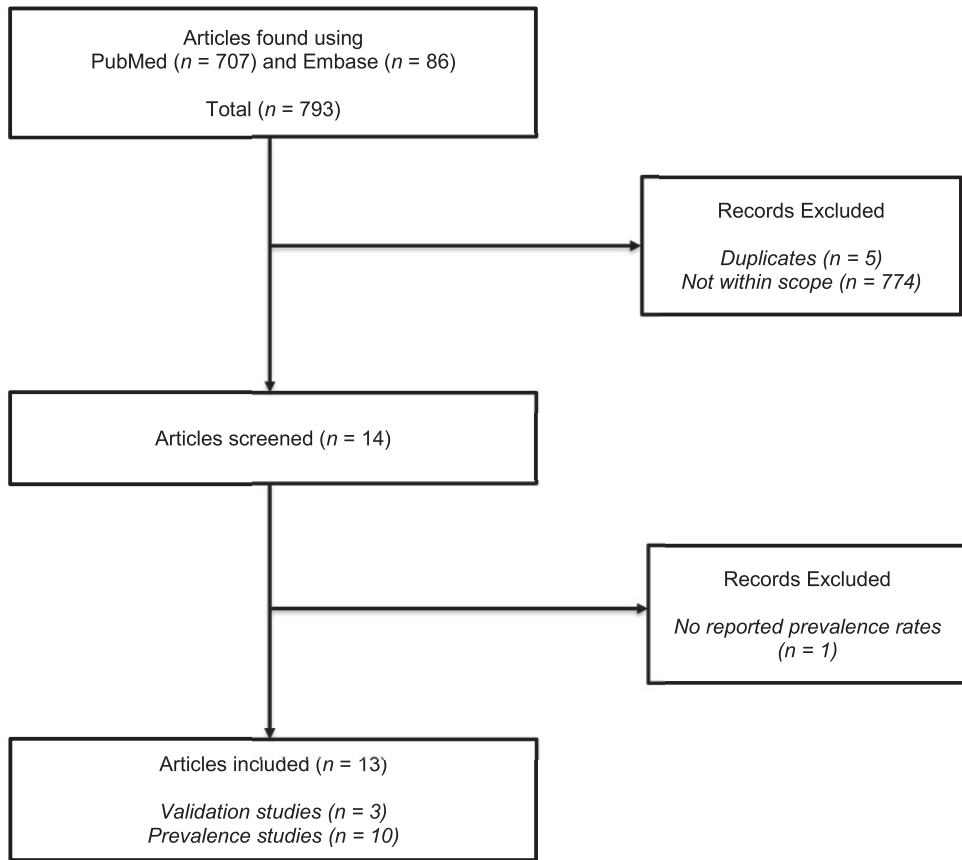


Figure 1. Flowchart depicting the screening process following a database search and literature review for studies that investigated neuropsychiatric symptoms in patients with Parkinson's disease from Arab countries.

or tools used for neuropsychiatric-symptom assessment, and was excluded.

Depression, anxiety, and apathy

Prevalence rates ranged between 22 and 76.7% for depression (seven studies),^{7,17–22} and 30–50.9% for anxiety (three studies).^{18,20,21}

Three Egyptian studies reported a prevalence of 85–87% for mood and cognitive symptoms combined using the NMSS, domain 3 (six questions).^{7,22,23} Khedr et al.,²² also reported sadness (41.1–72.7%), nervousness (60–63.6%), and lack

of motivation (32.2–63.6%) using the self-reported questionnaire for non-motor symptoms (NMSQuest).

Depression was assessed using the following validated Arabic screening tools: the Beck's Depression Inventory (BDI, $n = 1$), Hamilton Depression Rating Scale (HAM-D, $n = 2$), Montgomery–Asberg Depression Rating Scale (MADRS, $n = 1$), and Geriatric Depression Scale (GDS, $n = 1$). Three studies used the Diagnostic and Statistical Manual of mental disorders, fourth edition (DSM-IV) diagnostic criteria for a major depressive episode. Anxiety was assessed using the Hamilton Anxiety Rating

Table 1. Studies validating psychometric tools in Arab patients diagnosed with Parkinson's disease, presented in chronological order.

Study	Tool validated	Countries involved	Number of patients	Psychometric properties
Khalil et al., 2022 ¹³	MDS-UPDRS	Jordan Saudi Arabia Tunisia Egypt Kuwait Bahrain Palestine UAE	441	Factor structure consistent with English version (comparative fit index of ≥ 0.9)
Sellami et al., 2016 ¹⁴	NMSS	Tunisia	62	Cronbach's alpha = 0.87
Magdy et al., 2023 ¹⁵	PDSS-2	Egypt	133	Cronbach's alpha = 0.89 Sensitivity = 91% Specificity = 70%

MDS-UPDRS, Motor Disorder Society Unified Parkinson's Disease Rating Scale; NMSS, Non-Motor Symptom Scale; PDSS, Parkinson's Disease Sleep Scale.

Scale (HAM-A, $n=2$) and NMSQuest ($n=1$) as screening tools, with one study using the DSM-IV diagnostic criteria concomitantly with HAM-A.

Only one Egyptian study assessed the prevalence of apathy as an isolated construct, using the 14-item Apathy Scale (AS) in a sample of 50 patients. The reported prevalence was 70% using a cut-off score of 14 out of a maximum of 42.¹⁹ Importantly, while an Arabic version of the AS was used, the scale has no validated version in Arabic and the authors did not mention how the scale was translated.

Cognitive impairment

A total of five studies (all conducted in Egypt) reported the prevalence of cognitive impairment in their samples. Only two utilized the Montreal Cognitive Assessment (MOCA) screening tool. The first reported a 26% prevalence of cognitive decline in 41 patients who scored between 16 and 23,²⁰ whereas the second reported a much higher prevalence of 76% in a sample of 50 patients, with 25 scoring between 18 and 25, indicating mild impairment, and

nine scoring <18 , indicating more severe dementia.¹⁹

As mentioned above, the remaining three studies co-reported mood and cognitive symptoms using the NMSS and NMSQuest. Nevertheless, the prevalence of attention and memory symptoms were found to be approximately 72.5% using the NMSS, domain 5 (3 three questions).^{7,22,23} Among the three studies using the NMSS, only one study ran the Mini-Mental State Examination (MMSE) cognitive screener in parallel, and found a prevalence of 22.3% for dementia (mostly mild, MMSE score 19–21).²²

Perceptual disturbances or psychosis

A total of four studies reported prevalence rates of perceptual disturbances (hallucinations and related psychotic phenomena), with a range between 3 and 37%.^{7,20,22,23} Three studies used the NMSS, while one study used the Scale for Evaluation of Neuropsychiatric Disorders in Parkinson's Disease (SEND-PD), which is not validated in Arabic.

Delusions were explicitly mentioned in two studies, with a prevalence range of

Table 2. Prevalence studies of neuropsychiatric symptoms in Arab patients diagnosed with Parkinson’s disease (PD), presented in chronological order.

Study	Country	Sample	Participant characteristics	Duration of PD, years	PD stage	Prevalence of neuropsychiatric symptoms assessed	Instruments and questionnaires	Translated and validated
Zein and Mirghani, 2007 ²⁵	Sharjah, UAE	50	Patients in the Neurology clinic at Al-Qassimi Hospital Mean age, 71.1 years	7.8	Not mentioned	Dementia (26%) Depression (22%) Psychosis (10%) Visual hallucinations (6%) Sleep disorders (6%)	MMSE GDS (version unknown) Direct Interviewing Direct Interviewing Direct Interviewing	Yes Yes
Khedr et al., 2012 ²³	Assuit, Egypt	33	Community-based study with representative random sampling of 7 districts in the Assiut Governorate, Egypt Mean age, 66.94 ± 8.45 years	6.17 ± 5.9	Stage 1 (30.4%) Stage 2 (42.4%) Stage 3 (12.2%) Stage 4–5 (15%)	Mood/Cognition (84.8%) Attention/Memory (72.7%) Perceptual problems (3%) Sleep/Fatigue (84.8%)	NMSS NMSS NMSS NMSS	Yes Yes Yes Yes
Khedr et al., 2013 ²²	Assuit, Egypt	112	12 Females Outpatients in Assiut University Hospital Mean age, 61.0 ± 12.7 years	6.2 ± 5.9	Stage 1 (37.5%) Stage 2 (42.8%) Stages 3–5 (19.7%)	Mood/Cognition (87.5%) Attention/Memory (72.3%) Dementia (22.3%) Perceptual (9.9%) Sleep/Fatigue (78.6%) Depression (46%)	NMSS NMSS MMSE NMSS NMSS DSM-IV diagnostic criteria	Yes Yes Yes Yes Yes N/A
Ghaddar et al., 2016 ¹⁷	Beirut, Lebanon	200	35 Females Parkinson, Memory and Movement Disorders Center (Data from outpatients’ files) 80% Lebanese Mean age, 72.8 years	Not specified	Mild stage (34%) Moderate stage (48%) Severe stage (16%)			
Shalash et al., 2018 ⁷	Cairo, Egypt	97	76 Females Outpatients in Ain El Shams University Hospitals Mean age, 55.3 ± 10.8 years 39 Females	5.3 ± 4.1	Stage 1 (6.4%) Stage 1.5 (12.8%) Stage 2 (13.8%) Stage 2.5 (11.7%) Stage 3 (34%) Stage 4 (12.8%) Stage 5 (8.5%)	Mood/Cognition (87%) Depression (76.7%) Attention/Memory (72.4%) Sleep/Fatigue (91.3%) Perceptual problems (37%)	NMSS BDI NMSS NMSS NMSS	Yes Yes Yes Yes Yes

(continued)

Table 2. Continued.

Study	Country	Sample	Participant characteristics	Duration of PD, years	PD stage	Prevalence of neuropsychiatric symptoms assessed	Instruments and questionnaires	Translated and validated
Tibar et al., 2018 ²¹	Rabat, Morocco	117	Different regions of Morocco in the Department of Neurology and Neurogenetics at Ibn Sina University Hospital of Rabat Mean age, 60.77 ± 11.36 years	6	Stage < 2 (72.6%)	Depression (47.9%) Anxiety (50.9%) Sleep (80.6%) Daytime sleepiness (23.4%)	MADRS HAM-A PSQI ESS	Yes Yes Yes Yes
El Otmani, 2019 ²⁴	Casablanca, Morocco	125	Patients recruited during their consultation appointment Patients without memory impairment who were treated with dopamine agonists for ≥ 6 months Mean age, 57 ± 7 years 71 Females	ICD (+) = 7 ICD (-) = 8	Assessed motor severity but did not report them	ICDs (28%) At least two ICDs were found in 14% of patients	QUIP-RS	No
Ragab et al., 2019 ²⁰	Tanta, Egypt	41	Outpatients in Tanta University Hospitals Mean age, 57.95 ± 11.94 years 22 Females	2.7 ± 2.08	Stage 1 (51.2%) Stage 2 (26.8%) Stage 3 (22%)	Depression (47.5%) Dementia (26%) Sleep disturbance (36.6%) RLS (4.8%) Anxiety (30%) Psychosis (9.7%)	NMSQuest MOCA PSQI IRLS NMSQuest SEND-PD	Yes Yes Yes Yes Yes No
Khedr et al., 2020 ¹⁸	Egypt	64	Outpatients in South Valley University Hospital, Assiut University Hospital, and Aswan University Hospital	5.8 ± 3.2	Stage 1.5 (10.9%) Stage 2.5 (32.8%) Stage 3 (42.2%) Stage 4 (14.1%)	Depression (31.25%) Anxiety (40.6%) Overlap between depression and anxiety (23.4%)	DSM-IV criteria DSM-IV criteria (Also used HAM-D & HAM-A for scoring)	N/A

(continued)

Table 2. Continued.

Study	Country	Sample	Participant characteristics	Duration of PD, years	PD stage	Prevalence of neuropsychiatric symptoms assessed	Instruments and questionnaires	Translated and validated
Shaheen et al., 2022 ¹⁹	Cairo, Egypt	50	Outpatients with idiopathic PD in a Movement Disorders clinic at Cairo University Hospital Mean age, 62.88 ± 8.74 years 15 Females (30%)	3.76 ± 3.16	Mild (38%) Moderate (42%) Severe (20%)	Apathy (70%) Cognitive impairment (76%) Depression (68%) Sleep disturbances (76%)	AS MOCA HAM-D PDSS	No Yes Yes Yes

AS, Apathy Scale; BDI, Beck Depression Inventory; DSM-IV, Diagnostic and Statistical Manual, 4th edition; GDS, Geriatric Depression Scale; HAM-A, Hamilton Anxiety Rating Scale; HAM-D, Hamilton Depression Rating Scale; ESS, Epworth Sleepiness Scale; IRLS, International Restless Legs Study Group Severity Rating Scale; MADRS, Montgomery-Asberg Depression Rating Scale; MMSE, Mini-Mental State Examination; MOCA, Montreal Cognitive Assessment; NMSQuest, Non-Motor Symptom Questionnaire; NMSS, Non-Motor Symptom Scale; PDSS, Parkinson Disease Sleep Scale; PSQI, Pittsburgh Sleep Quality Index; QUIP-RS, Questionnaire for Impulsive-Compulsive Disorders in Parkinson's Disease Rating Scale; SEND-PD, Scale for Evaluation of Neuropsychiatric Disorders in Parkinson's Disease; ICD, impulse control disorders.

9.8–14.1%,^{7,22} assessed separately using item 14 of the NMSS, asking if the patient holds beliefs that the investigator knows are not true.

Impulse control disorders (ICDs)

Only one Moroccan study assessed the prevalence of ICDs using a direct translation of the Questionnaire for Impulsive-Compulsive Disorders in Parkinson's Disease Rating Scale (QUIP-RS) in a sample of 125 patients with PD.²⁴ The overall prevalence of ICDs was reported to be 28%, with 14% of patients exhibiting at least two ICDs. Prevalence rates of specific ICDs were identified, including pathological gambling (3.2%), compulsive sexual behavior (7.2%), pathological buying (9.6%), eating behavior disorder (7.2%), and punding/hobbyism (11.1%). Culturally specific ICD-like behaviors including 'excessive charity' (18.4%) and excessive reading of the Qur'an (9.6%) were excluded.

Sleep disorders

A total of seven studies investigated sleep disorders in Arab patients with PD. Three studies grouped sleep disturbances and fatigue, using the NMSS, domain 2 (four questions). The authors reported their joint prevalences as 78.6%,²² 84.8%,²³ and 91.3%,⁷ respectively.

One Egyptian study used the PDSS and reported sleep disturbances in 76% of the sample.¹⁹ An Emirati study found a 6% prevalence of sleep disorders using direct interviewing by asking about vivid dreams, sleep talking, and sleep movements or falls during sleep.²⁵ Two studies used the Pittsburgh Sleep Quality Index (PSQI), with a 36.6% prevalence of sleep disturbances in the Egyptian study (including 4.8% who had restless leg syndrome),²⁰ and 80.6% in the Moroccan study.²¹ The latter also

reported daytime sleepiness as 23.4% using the Epworth Sleepiness Scale (ESS).

Validation studies in patients with PD

The multisite validation study conducted in Jordan, Saudi Arabia, Tunisia, Egypt, Kuwait, Bahrain, Palestine, and UAE, created and validated the Arabic version of the MDS-UPDRS, which was adopted as the official version by the Movement Disorder Society. The MDS-UPDRS is a widely used measure and is considered the gold-standard global clinical assessment of patients with PD. The questionnaire was first translated through forward and backward translation by Khalil et al.,¹³ and received approval from the Movement Disorder Society Task Force. The questionnaire was then validated in a convenience sample of 441 patients with PD, recruited from 14 different sites in the above-mentioned eight Arab countries. Of note, the MDS-UPDRS evaluates all domains of PD (motor and non-motor) including neuropsychiatric symptoms.¹³ Limitations of the aforementioned study are related to most patients having mild-moderate stage PD severity, and a lack of representativity from all 22 Arab countries. This tool is copyrighted and requires permission to be used in clinical and research settings.

A Tunisian study translated and validated the Arabic version of the NMSS in 62 patients in Razi Hospital.¹⁴ The NMSS evaluates the following domains: sleep/fatigue, mood/cognition, perceptual disturbances/hallucinations, attention/memory, and autonomic dysfunction, among others, with a Cronbach's alpha of 0.87.¹⁶

A study conducted in Egypt validated the PDSS-2 in 133 patients with PD attending two neurology outpatient clinics. Cronbach's alpha was 0.89, with a sensitivity of 91% and specificity of 70% in differentiating good from poor sleepers, using a cut-off score of 13 over a total of 60.¹⁵

Discussion

The present article is the first to review existing studies regarding the assessment and prevalence of neuropsychiatric symptoms in Parkinson's disease in the Arab world, and showcases the scarcity of relevant studies, with Egypt being involved in eight out of the 13 studies identified. The present article also highlights the methods and scales used to report the prevalence of neuropsychiatric symptoms, leading to heterogeneous findings in the Arab population.

Epidemiological studies

Most studies included in the present review were conducted in tertiary care centers and no longitudinal or incidence studies were conducted.

Depression. In general, depression in PD varies in frequency and severity with a mean 35% prevalence of clinically significant depressive symptoms, and a 17% prevalence of a major depressive disorder (MDD).²⁶ Importantly, many symptoms, such as apathy and pseudobulbar affect, may complicate the diagnosis of mood disorders in PD. The GDS is an adequately reliable instrument that was found to be less confounded by somatic symptoms associated with PD. Compared with other scales, such as the BDI, MADRS, and HAM-D, research suggest that the GDS might be a more sensitive screening tool for depression in patients with PD.²⁷

Out of the 10 studies included in the present review, only one used the recommended GDS,²⁵ with a reported depression prevalence of 22%. Unfortunately, the authors didn't specify which version of the GDS they used (the 15 or 30-item tool). The other studies found prevalence rates higher than 45%, which may be attributed to the use of less reliable tools in PD, such as the BDI,⁷ MADRS,²¹ HAM-D,¹⁹ or non-specific

general screening tools, such as the NMSS or NMSQuest.^{7,20,22,23}

Both Kheder et al.²² and Ghaddar et al.¹⁷ reported higher prevalence rates of MDD compared with the available literature (31% and 46%, respectively, versus 17%), using the gold standard DSM-IV diagnostic criteria, which is an alarming finding in Arab-speaking populations. This finding might be explained by the fact that more than half of the recruited patients had at least moderate stages of PD. In fact, while depression may often be seen at disease onset, its prevalence increases as the disease progresses.^{6,28}

Anxiety. Like depression, anxiety is often seen at the onset of PD. However, some studies report a decrease in anxiety prevalence with disease progression.^{6,28,29} The estimated prevalence of anxiety disorders in Western populations with PD is 31%, with generalized anxiety disorder being the most common disorder.³⁰ Patients with PD experience internal tremor and social or anticipatory anxiety more often than the general population. Consequently, standard anxiety scales that account for physical symptoms, including the HADS, HAM-A, and Beck's Anxiety Inventory, may not be the most appropriate screening tools in patients with PD, as the ability of these scales to discriminate somatic symptoms between anxious and non-anxious patients is markedly compromised in PD.³¹

Many instruments have been devised to detect anxiety in PD. The Geriatric Anxiety Inventory (GAI) showed the best balance between specificity (0.88) and sensitivity (0.86).³² The Parkinson's Anxiety Scale (PAS) was also specifically developed for this aim, as it minimizes the assessment of somatic anxiety symptoms that might be confused with PD markers.³³ Although slightly superior in specificity (0.91) to the GAI, the PAS has markedly lower sensitivity (0.71).³² Unfortunately, none of these

scales have been translated or validated in Arabic. The studies included in the present review demonstrated much higher estimates of anxiety compared with the abovementioned estimates, ranging between 30 and 50%, using suboptimal tools including the NMSQuest,²⁰ and HAM-A.²¹ The only study that used a diagnostic tool (the DSM-IV) to assess for anxiety, showed slightly higher estimates of generalized anxiety disorder (40%) compared with the Western literature,¹⁸ which highlights a possible higher prevalence of anxiety disorders in Arab-speaking populations.

Apathy. Apathy is a multidimensional construct that can be defined as decreased motivation associated with a decrease in goal-directed activity and emotional indifference. A meta-analysis by Den Brok et al.³⁴ found the mean prevalence of apathy in PD to be around 40%. However, the included studies were considerably heterogeneous and exclusively conducted on Western populations.³⁴ A more recent systematic review, including seven Western studies, found the apathy rate to be approximately 30%.³⁵ The Lille Apathy Rating Scale (LARS) has been shown to reliably distinguish between depression and apathy, with sensitivity and specificity rates of 89 and 92%, respectively,³⁵ leading to the superiority of this tool in terms of clinical validity in patients with PD.³⁶ Unfortunately, none of the Arab studies included in the present review used the LARS, which is not translated or validated in Arabic. The only study using a specific screening tool, the AS, reported a relatively high 70% apathy prevalence rate, which is close to the depression rate of 68% reported in the same population studied. It is noteworthy that the AS used in this study was not validated in Arabic.¹⁹

Cognitive impairment. Cognitive impairment often manifests early in individuals with

PD. In fact, approximately 20–42% of patients newly diagnosed with PD are shown to have mild cognitive impairment.³⁷ When these patients are followed-up after 5-years, 11–28% may have returned to normal cognition, while 39–50% will eventually develop dementia.³⁸ The most commonly used methods to assess cognitive impairment in PD are the MOCA and MMSE.³⁹ Nonetheless, the MOCA has shown superiority over the MMSE in detecting mild cognitive impairment in PD.⁴⁰ In the present review, only two studies used the MOCA as a screening tool, showing a wide variability in prevalence range (between 27 and 70%), which might be explained by differences in severity and duration of PD in the included patients.^{19,20} The rest of the studies were limited by using the NMSS that combines mood and cognitive symptoms and separates the latter from attention and memory problems.^{7,22,23,25} In fact, when one study ran the MMSE in parallel with the NMSS, the difference in estimates was vast, from 87.5% (mood/cognition) and 72.3% (attention/memory) on NMSS to only 22.3% on the MMSE.²² While both the MMSE and MOCA are validated and widely used in Arab-speaking populations,^{41,42} they seem underused in the Arab subpopulation with PD. In addition, the Scales for Outcomes in Parkinson's Disease – Cognition (SCOPA-COG) is a short tool to assessing four cognitive domains that has been shown to be more discriminant to smaller changes than the MMSE.⁴³ However, this scale has not been translated into Arabic.

Perceptual disturbances. Hallucinations affect 42% of patients with PD and are more common as the disease advances.^{6,44} Minor hallucinatory phenomena, encompassing presence hallucinations (feeling that somebody is nearby), passage hallucinations (brief visual hallucinations), and visual illusions (misperception of real

objects), are frequent (42%) early premotor symptoms.⁴⁵ Delusions are less common with a range of 3–14%, and are most commonly paranoid delusions related to infidelity.⁴⁶ The diagnosis of PD psychosis is becoming increasingly based on the National Institute of Neurological Disorders and Stroke and National Institute of Mental Health (NINDS-NIMH) framework, which does not require the symptoms to be distressing.⁴⁶ To date, no scale is specifically recommended to assess for psychosis in PD, however, the Scale for Assessment of Positive Symptoms Parkinson's Disease (SAPS-PD) and the Neuropsychiatric Inventory (NPI) are considered clinically useful tools.^{46,47} The SEND-PD is also useful for evaluating psychosis, as well as mood/apathy and ICDs.⁴⁸ While the NPI is validated in Arabic, the SEND-PD and SAPS-PD are not.⁴⁹ The Parkinson Psychosis Rating Scale (PPRS) is another tool that was initially designed to assess the severity of psychotic symptoms induced by dopaminergic medications in patients with PD.⁵⁰ Unfortunately, this tool has been underused in clinical studies in patients with PD, as it was not shown to be superior to the aforementioned assessment scales.^{47,51} Three studies in the present review used the NMSS,^{7,22,23} which is known to have low internal consistency and high floor effect for the perceptual domain.⁵² Heterogeneity was evident with rates ranging from 3% to 37%, although it is worth mentioning that disease stage differed between samples. A fourth study did not use any tool but estimated the prevalence of psychosis through direct interviewing, and did not mention which diagnostic criteria they relied on.²⁵ The final study used a non-validated version of the SEND-PD.²⁰ Hence, not a single study included in the present review assessed perceptual disturbances using an optimal tool.

Impulse control disorders. Impulse control disorders are not only a product of PD itself but are often the byproduct of dopaminergic treatments of PD, notably dopaminergic agonists.^{6,53} For instance, two studies found an increase from baseline rates of ICDs in PD (about 20%) to more than 30% after 5 years of treatment.^{54,55} Limited tools are available to assess for ICDs in PD. The Minnesota Impulsive Disorders Interview (MIDI) is one tool that covers a range of ICDs, however, its limitations include being too broad and not user-friendly.^{56,57} The QUIP-RS was adopted by the Movement Disorder Society as a practical, sensitive, and specific tool ($\geq 80\%$) to assess gambling, buying, sex, eating, and hobbyism and punting in patients with PD.⁵⁶ ICDs in PD are particularly understudied in the Arab world, with only one study documenting a prevalence of 28%.²⁴ Unfortunately, the study used a non-validated version of the QUIP-RS, excluded those with memory impairment, and did not correlate findings with dopaminergic medication use.

Sleep disturbances. Even though sleep problems are experienced by four out of five patients with PD, they remain underdiagnosed clinically.⁵⁸ Sleep problems appear in 60% of patients with PD as insomnia,⁵⁹ in approximately 50% of patients as rapid eye movement sleep behavior disorder (RBD),⁵⁸ in 15–46% of patients as daytime sleepiness,⁵⁸ and in 8–24% of patients as restless leg syndrome (RLS),⁵⁸ among others. Most studies in the present review did not categorize the investigated sleep disorders. Studies using the NMSS overestimated the prevalence ($>75\%$), likely due to inclusion of the fatigue domain and the ill-definition of sleep disturbances.^{7,22,23} Shaheen et al.¹⁹ found a similar prevalence (76%) when using the PDSS. The PDSS, while a useful tool to assess for insomnia, is not as robust indicator of other sleep

disorders, which led to the development of the PDSS-2.^{60,61} Interestingly, the two studies that used the PSQI in the present review found two differing prevalences (36.6% and 80.6%, respectively),^{20,21} although most included patients in both studies had similar severity of PD ($\leq$ stage 2). This difference may be attributed to differing sociocultural factors between the two countries (e.g. Morocco has less noise pollution compared with Egypt),⁶² differing sample size, as well as differing characteristics of the included patients (with or without cognitive impairment).

Assessment tools for sleep disorders are inherently limited by the number of sleep disorders, patient's sleep perception, and overlap in symptoms, among others.⁶³ In a review published in the Movement Disorder Society journal, the PDSS, PSQI, Scale for Outcomes in Parkinson's Disease – Sleep (SCOPA-S), and ESS, emerged as recommended tools assessing for different aspects of sleep disorders.⁶³ For instance, the SCOPA-S assesses both excessive daytime sleepiness and problems with night sleep.⁶¹ Alternatively, the PSQI can be used to assess overall sleep impairment and the ESS to assess daytime sleepiness.⁶³ The PDSS-2 (but not the original PDSS),¹⁵ PSQI,⁶⁴ and ESS,⁶⁵ are validated in Arabic, while the SCOPA-S is not.

Finally, RBD can be screened for using the REM Sleep Behavior Disorder Screening Questionnaire (RBDSQ),⁶⁶ and although a single screening question (RBD1Q) has shown promise with high sensitivity and specificity,⁶⁷ no validated version exists in Arabic. For RLS, The Movement Disorder Society recommends assessment using the International Restless Legs Scale (IRLS),⁶¹ which is validated in Arabic.⁶⁸ Thus, we recommend combining short but reliable tools such as the PDSS, PSQI, SCOPA and the ESS to improve the detection of specific sleep disorders and guide treatment in patients with PD.

Challenges in validation of neuropsychiatric symptom assessment tools across Arab countries

Although the present review sheds light on the validation of three specific instruments (the NMSS, MDS-UPDRS and the PDSS-2) for assessment of neuropsychiatric symptoms in patients with PD, these instruments are not without limitations. It is important to recognize the need to culturally validate Western tools, as certain symptoms are preferentially expressed in different populations, such as somatic and pain symptoms in Arab populations.⁶⁹ Differences also exist within Arab countries, most importantly the inherent variability of the Arabic dialects, which can sometimes be mutually unintelligible. Currently, six major groups of dialects are recognized.⁷⁰ The variability of dialects is particularly relevant for non-self-report tools (e.g. clinician-administered) translated into written Arabic, known as Modern Standard Arabic (MSA). The latter, almost equivalent to contemporary Latin, is almost never spoken and would not be appropriate in verbal settings, such as an interview.⁷¹

The NMSS was translated into MSA and validated in 62 Tunisian patients, which undermines the generalizability of this tool due to the very limited sample size. Moreover, the authors reported difficulties encountered with individuals who were not very familiar with MSA, who struggled with understanding the language without outside explanation. Beside the limitations of translation, there are also limitations inherent to the tool itself. While monumental at the time of its inception, the Movement Disorder Society itself recognizes the limitation of the grouping of certain symptoms (e.g. sleep disorders with fatigue), and poor assessment of other domains, such as cognition and impulse control disorders. These limitations have led to the development of the International Parkinson and Movement Disorder Society Nonmotor Rating Scale

(MDS-NMS).^{52,72} At the time of the present review, the validation of an Arabic version is still underway.

The International Parkinson and Movement Disorder Society Unified Parkinson's Disease Rating Scale (MDS-UPDRS) has become the gold standard for evaluation of multiple key aspects of PD in clinical and research settings.⁵¹ The Arabic version demonstrates good validity and utility in Arabic-speaking patients with PD and has been thereby designated as an official MDS-UPDRS version.¹³ The MDS-UPDRS individual items were shown to have strong correlations ($r > 0.60$) with the LARS that specifically assesses for apathy in PD, the PPRS that specifically assesses for psychosis in PD, and PSQI that specifically assesses for overall sleep disorders in patients with PD. However, the MDS-UPDRS individual items have low to moderate correlation with HAM-D and A scales ($r = 0.40$ – 0.60) and show unsatisfactory correlations for three cognitive scales ($r < 0.40$).⁷³ Thus, in future studies, we recommend combining the MDS-UPDRS part I screening tool with more specific tools, notably for depression, anxiety, and cognitive impairment domains, to better ascertain prevalence rates. We also recommend translating and validating specific assessment tools for PD, namely the LARS and the GAI for a more accurate assessment of apathy and anxiety, respectively, in patients with PD. Importantly, due to the overlap of symptoms in PD, tools should be validated against clinician diagnoses.

The best recommended tools to be used in patients with PD, according to the international literature, are summarized in Table 3, along with the status of their validation in Arabic.

Common limitations of all studies

First, out of the 22 Arab countries, only four (Egypt, Lebanon, Morocco, and the

Table 3. Recommended assessment scales for neuropsychiatric symptoms in patients with Parkinson's disease.

Recommended scale	Recommended neuropsychiatric symptom domain	Validated in Arabic	Used in PD in the Arab World?	Comments
MDS-UPDRS	All	Yes	Yes	Also assesses for motor symptoms
MDS-NMS	All	Underway	No	The revision of the NMSS
NMSS	All	Yes	Yes ($n = 3$)	Has been revised into MDS-NMS
GDS	Depression	Yes	Yes ($n = 1$)	
GAI	Anxiety	No	No	
MOCA	Cognition	Yes	Yes ($n = 2$)	
MMSE	Cognition	Yes	Yes ($n = 1$)	
AS	Apathy	No	Yes ($n = 1$)	
LARS	Apathy	No	No	
SAPS-PD	Psychosis	No	No	
NPI	Psychosis	Yes	No	Can assess cognitively impaired
SEND-PD	Psychosis	No	No	Also assessed other neuropsychiatric symptoms
QUIP-RS	ICDs	No	Yes ($n = 1$)	
PDSS	Sleep Disorders	No	Yes ($n = 1$)	Unsuitable for assessment of RLS, RBD, sleep apnea, and EDS
PDSS-2	Sleep disorders	Yes	No	Assessed wider spectrum of sleep disorders than original PDSS
SCOPA-S	Sleep quality and EDS	No	No	
PSQI	Sleep quality	Yes	Yes ($n = 2$)	
ESS	EDS	Yes	Yes ($n = 1$)	
IRLS	RLS	Yes	Yes ($n = 1$)	
RBDSQ	RBD	No	No	RBD IQ is promising

AS, Apathy Scale; EDS, Excessive Daytime Sleepiness; ESS, Epworth Sleepiness Scale; GAI, Geriatric Anxiety Inventory; GDS, Geriatric Depression Scale; ICDs, Impulse Control Disorders; IRLS, International Restless Legs Study Group Severity Rating Scale; LARS, Lille Apathy Rating Scale; MDS-NMS, International Parkinson and Movement Disorder Society Nonmotor Rating Scale; MDS-UPDRS, International Parkinson and Movement Disorder Society Unified Parkinson's Disease Rating Scale; MMSE, Mini-Mental State Examination; MOCA, Montreal Cognitive Assessment; NPI, Neuropsychiatric Inventory; NMSS, Non-Motor Symptom Scale; PD, Parkinson's Disease; PDSS, Parkinson Disease Sleep Scale; PSQI, Pittsburgh Sleep Quality Index; QUIP-RS, Questionnaire for Impulsive-Compulsive Disorders in Parkinson's Disease Rating Scale; RBD, REM Sleep Behavior Disorder; RBDSQ, REM Sleep Behavior Disorder Screening Questionnaire; RLS, Restless Leg Syndrome; SAPS-PD, Scale for Assessment of Positive Symptoms Parkinson's Disease; SCOPA-S, Scale for Outcomes in Parkinson's Disease – Sleep; SEND-PD, Scale for Evaluation of Neuropsychiatric Disorders in Parkinson's Disease.

UAE) contributed to the 10 prevalence studies, and over half of these studies were conducted in Egypt. Secondly, the sample sizes are limited, with no study comprising

more than 200 patients, and six out of 10 studies having less than 100 patients. Thirdly, there was a heterogeneity in the duration of PD in the included patients,

ranging between 2.7 and 8 years, while the average duration of illness was approximately 10 years. However, this range is highly variable and depends on factors such as early treatment, age of onset, and motor subtype.⁷⁴ Although staging of PD severity was included as a variable in many studies, most included patients with mild–moderate disease, and some studies excluded severely ill patients, and those with cognitive impairment, based on certain cut-offs on cognitive screening tests. This is particularly important as neuropsychiatric symptoms, namely depression, apathy, psychosis, and cognitive impairment, have increasing prevalence as PD progresses in terms of duration and severity.²⁸ Of note was the heavy reliance on symptom rating tools rather than clinical assessment based on diagnostic criteria (except in two studies), which may have overestimated the prevalence.⁷⁵ Indeed, certain studies used only screening tools, such as the BDI, to estimate prevalence, which would overestimate the results. Future studies using diagnostic assessments of neuropsychiatric symptoms are thus warranted. Finally, all studies did not account for concurrent neuropsychiatric symptom treatment with pharmacotherapy (e.g., antidepressants) or psychotherapy. Patients on therapy were either excluded or the difference in prevalence of neuropsychiatric symptoms between medicated groups and non-medicated groups was not analyzed.

Overall, the available literature on neuropsychiatric symptoms in Arab-speaking patients with PD is poorly representative of the general population of patients with PD in the countries where studies have been conducted, let alone the Arab world. Thus, researchers from all different Arab countries should join efforts to increase their research sample size, and generalizability of findings to Arab-speaking populations. Future studies are recommended to include patients across all stages of PD to increase

the generalizability of findings and to control for the impact of disease duration and severity, as well as use of antiparkinsonian drugs, psychotropics and psychotherapy on the emergence and severity of neuropsychiatric symptoms in this population.

A final limitation of the present review is the fact that the search was restricted to PubMed and Embase as bibliographic databases.

Conclusion

The present article provides a comprehensive overview of studies conducted in the 22 Arab countries assessing for neuropsychiatric symptoms including depression, anxiety, apathy, cognitive impairment, psychosis, impulse control, and sleep disorders in patients diagnosed with PD. The review also summarizes the tools that were used, whether validated or not. The authors compared their findings with the international literature, and identified methodological limitations in the studies included, in addition to shortcomings of the assessment tools used. The authors provided recommendations to strengthen the evidence base behind future studies exploring the prevalence and severity of neuropsychiatric symptoms in patients with PD in the Arab-speaking countries.

Author contributions

AC, HK, CA, and AS contributed to the literature review and writing different sections of the paper. RK conceptualized and designed the research, and critically reviewed the different drafts written.

Data availability

Data sharing is not applicable to this article as all relevant information is contained within this review.

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ORCID iDs

Antoine Sader  <https://orcid.org/0009-0004-3405-2746>

Rita Khoury  <https://orcid.org/0000-0003-2481-6816>

References

1. Kalia LV and Lang AE. Parkinson's disease. *Lancet* 2015; 386: 896–912.
2. Faddoul L, Chahine B, Haydar S, et al. The effect of pramipexole extended release on the levodopa equivalent daily dose in Lebanese Parkinson diseased patients. *Pharm Pract (Granada)* 2018; 16: 1220.
3. Dorsey ER, Sherer T, Okun MS, et al. The emerging evidence of the Parkinson pandemic. *J Parkinsons Dis* 2018; 8: S3–S8.
4. Safiri S, Noori M, Nejadghaderi SA, et al. The burden of Parkinson's disease in the Middle East and North Africa region, 1990–2019: results from the global burden of disease study 2019. *BMC Public Health* 2023; 23: 107.
5. He R, Zhao Y, He Y, et al. Olfactory dysfunction predicts disease progression in Parkinson's disease: a longitudinal study. *Front Neurosci* 2020; 14: 569777.
6. Weintraub D, Aarsland D, Chaudhuri KR, et al. The neuropsychiatry of Parkinson's disease: advances and challenges. *Lancet Neurol* 2022; 21: 89–102.
7. Shalash AS, Hamid E, Elrassas HH, et al. Non-motor symptoms as predictors of quality of life in Egyptian patients with Parkinson's disease: a cross-sectional study using a culturally adapted 39-item Parkinson's disease questionnaire. *Front Neurol* 2018; 9: 357.
8. League of Arab States: Monitoring Arab efforts to eradicate multidimensional poverty. *Arab Strategic Framework for the Eradication of Multidimensional Poverty* 2020–2030.
9. Arab American Anti-Discrimination Committee (ADC). *Facts about Arabs and the Arab World*, <https://adc.org/facts-abo-ut-arabs-and-the-arab-world/> (2009, accessed 13 June 2024).
10. World Bank Group. *Arab World*, <https://data.worldbank.org/region/middle-east-and-north-africa> (2022, accessed 13 June 2024).
11. EI Masri J, Dankar R, EI Masri D, et al. The Arab countries' contribution to the research of neurodegenerative disorders. *Cureus* 2021; 13: e17589.
12. Baethge C, Goldbeck-Wood S and Mertens S. SANRA—a scale for the quality assessment of narrative review articles. *Res Integr Peer Rev* 2019; 4: 5.
13. Khalil H, Aldaaani ZF, Aldughmi M, et al. Validation of the Arabic version of the Movement Disorder Society-Unified Parkinson's Disease Rating Scale. *Mov Disord* 2022; 37: 826–841.
14. Sellami L, Kacem I, Nasri A, et al. Evaluation of an Arabic version of the non-motor symptoms scale in Parkinson's disease. *Neurol Sci* 2016; 37: 963–968.
15. Magdy R, Mohammed Z, Hassan A, et al. Validation of the Arabic version of Parkinson's Disease Sleep Scale-Revised Version (PDSS-2). *Rev Neurol (Paris)* 2024; 180: 195–210.
16. International Parkinson and Movement Disorder Society. *Non-Motor Symptoms Scale for Parkinson's Disease (NMSS)*, <https://www.movementdisorders.org/MDS/MDS-Rating-Scales/Non-Motor-Symptoms-Scale-for-Parkinsons-Disease-NMSS.htm> (2007, accessed 16 July 2024).
17. Ghaddar A, Fawaz M, Khazen G, et al. Prevalence of depression in Parkinson's disease in a Lebanese tertiary clinic. *J Clin Exp Neuropsychol* 2016; 38: 51–58.
18. Khedr EM, Abdelrahman AA, Elserogy Y, et al. Depression and anxiety among patients with Parkinson's disease: frequency, risk factors, and impact on quality of life. *Egypt J Neurol Psychiatry Neurosurg* 2020; 56: 116.
19. Shaheen S, Ali RM, Farghaly M, et al. Screening for non-motor symptoms in Egyptian patients with Parkinson's disease. *Egypt J Neurol Psychiatry Neurosurg* 2022; 58: 103.

20. Ragab OA, Elheneedy YA and Bahnasy WS. Non-motor symptoms in newly diagnosed Parkinson's disease patients. *Egypt J Neurol Psychiatry Neurosurg* 2019; 55: 24.
21. Tibar H, El Bayad K, Bouhouche A, et al. Non-motor symptoms of Parkinson's disease and their impact on quality of life in a cohort of Moroccan patients. *Front Neurol* 2018; 9: 170.
22. Khedr EM, El Fetoh NA, Khalifa H, et al. Prevalence of non motor features in a cohort of Parkinson's disease patients. *Clin Neurol Neurosurg* 2013; 115: 673–677.
23. Khedr EM, Al Attar GS, Kandil MR, et al. Epidemiological study and clinical profile of Parkinson's disease in the Assiut Governorate, Egypt: a community-based study. *Neuroepidemiology* 2012; 38: 154–163.
24. El Otmani H, Mouni FZ, Abdulhakeem Z, et al. Impulse control disorders in Parkinson disease: a cross-sectional study in Morocco. *Rev Neurol (Paris)* 2019; 175: 233–237.
25. Zein TM and Mirghani Z. Mental changes in Parkinson's disease a study of fifty patients. *Folia Medica Indonesiana* 2007; 43: 28–31.
26. Reijnders JSAM, Ehrst U, Weber WEJ, et al. A systematic review of prevalence studies of depression in Parkinson's disease. *Mov Disord* 2008; 23: 183–189; quiz 313.
27. McDonald WM, Holtzheimer PE, Haber M, et al. Validity of the 30-item geriatric depression scale in patients with Parkinson's disease. *Mov Disord* 2006; 21: 1618–1622.
28. Dlay JK, Duncan GW, Khoo TK, et al. Progression of neuropsychiatric symptoms over time in an Incident Parkinson's Disease Cohort (ICICLE-PD). *Brain Sci* 2020; 10: 78.
29. Weintraub D, Caspell-Garcia C, Simuni T, Parkinson's Progression Markers Initiative, et al. Neuropsychiatric symptoms and cognitive abilities over the initial quinquennium of Parkinson disease. *Ann Clin Transl Neurol* 2020; 7: 449–461.
30. Broen MPG, Narayan NE, Kuijff ML, et al. Prevalence of anxiety in Parkinson's disease: a systematic review and meta-analysis. *Mov Disord* 2016; 31: 1125–1133.
31. Forjaz MJ, Martinez-Martin P, Dujardin K, et al. Rasch analysis of anxiety scales in Parkinson's disease. *J Psychosom Res* 2013; 74: 414–419.
32. Mele B, Holroyd-Leduc J, Smith EE, et al. Detecting anxiety in individuals with Parkinson disease: a systematic review. *Neurology* 2018; 90: e39–e47.
33. Leentjens AFG, Dujardin K, Pontone GM, et al. The Parkinson Anxiety Scale (PAS): development and validation of a new anxiety scale. *Mov Disord* 2014; 29: 1035–1043.
34. Den Brok MGHE, Van Dalen JW, Van Gool WA, et al. Apathy in Parkinson's disease: a systematic review and meta-analysis. *Mov Disord* 2015; 30: 759–769.
35. Mele B, Merrih D, Ismail Z, et al. Detecting apathy in individuals with Parkinson's disease: a systematic review. *J Parkinsons Dis* 2019; 9: 653–664.
36. Mohammad D, Ellis C, Rau A, et al. Psychometric properties of apathy scales in Parkinson's disease: a systematic review. *Neurodegener Dis Manag* 2018; 8: 267–282.
37. Aarsland D, Bronnick K, Williams-Gray C, et al. Mild cognitive impairment in Parkinson disease: a multicenter pooled analysis. *Neurology* 2010; 75: 1062–1069.
38. Domellöf ME, Ekman U, Forsgren L, et al. Cognitive function in the early phase of Parkinson's disease, a five-year follow-up. *Acta Neurol Scand* 2015; 132: 79–88.
39. Yu YW, Tan CH, Su HC, et al. A new instrument combines cognitive and social functioning items for detecting mild cognitive impairment and dementia in Parkinson's disease. *Front Aging Neurosci* 2022; 14: 913958.
40. Hoops S, Nazem S, Siderowf AD, et al. Validity of the MoCA and MMSE in the detection of MCI and dementia in Parkinson disease. *Neurology* 2009; 73: 1738–1745.
41. El-Hayek R, Baddoura R, Wehbé A, et al. An Arabic version of the Mini-Mental State Examination for the Lebanese population: reliability, validity, and normative data. *J Alzheimers Dis* 2019; 71: 525–540.
42. Rahman TT and El Gaafary MM. Montreal Cognitive Assessment Arabic version: reliability and validity prevalence of mild cognitive impairment among elderly attending

- geriatric clubs in Cairo. *Geriatr Gerontol Int* 2009; 9: 54–61.
43. Marinus J, Visser M, Verwey NA, et al. Assessment of cognition in Parkinson's disease. *Neurology* 2003; 61: 1222–1228.
 44. Fénelon G, Soulas T, Zenasni F, et al. The changing face of Parkinson's disease-associated psychosis: a cross-sectional study based on the new NINDS-NIMH criteria. *Mov Disord* 2010; 25: 763–766.
 45. Pagonabarraga J, Martínez-Horta S, Fernández de Bobadilla R, et al. Minor hallucinations occur in drug-naïve Parkinson's disease patients, even from the premotor phase. *Mov Disord* 2016; 31: 45–52.
 46. Schneider RB, Iourinets J and Richard IH. Parkinson's disease psychosis: presentation, diagnosis and management. *Neurodegener Dis Manag* 2017; 7: 365–376.
 47. Fernandez HH, Aarsland D, Fénelon G, et al. Scales to assess psychosis in Parkinson's disease: critique and recommendations. *Mov Disord* 2008; 23: 484–500.
 48. Rodríguez-Violante M, Cervantes-Arriaga A, Velázquez-Osuna S, et al. Independent validation of the SEND-PD and correlation with the MDS-UPDRS part IA. *Parkinsons Dis* 2014; 2014: 260485.
 49. Feghali Y, Fares Y and Abou Abbas L. Assessment of neuropsychiatric symptoms in dementia: validity and reliability of the Lebanese version of the neuropsychiatric inventory questionnaire. *Appl Neuropsychol Adult* 2021; 28: 588–595.
 50. Friedberg G, Zoldan J, Weizman A, et al. Parkinson Psychosis Rating Scale: a practical instrument for grading psychosis in Parkinson's disease. *Clin Neuropharmacol* 1998; 21: 280–284.
 51. Goetz CG. Scales to evaluate psychosis in Parkinson's disease. *Parkinsonism Relat Disord* 2009; 15: S38–S41.
 52. Van Wamelen DJ, Martínez-Martin P, Weintraub D, International Parkinson and Movement Disorder Society Parkinson's Disease Non-Motor Study Group, et al. The Non-Motor Symptoms Scale in Parkinson's disease: validation and use. *Acta Neurol Scand* 2021; 143: 3–12.
 53. Weintraub D, Koester J, Potenza MN, et al. Impulse control disorders in Parkinson disease: a cross-sectional study of 3090 patients. *Arch Neurol* 2010; 67: 589–595.
 54. Marković V, Stanković I, Petrović I, et al. Dynamics of impulsive-compulsive behaviors in early Parkinson's disease: a prospective study. *J Neurol* 2020; 267: 1127–1136.
 55. Corvol JC, Artaud F, Cormier-Dequaire F, DIGPD Study Group, et al. Longitudinal analysis of impulse control disorders in Parkinson disease. *Neurology* 2018; 91: e189–e201.
 56. Weintraub D, Mamikonyan E, Papay K, et al. Questionnaire for impulsive-compulsive disorders in Parkinson's Disease–Rating Scale. *Mov Disord* 2012; 27: 242–247.
 57. Christenson GA, Faber RJ, De Zwaan M, et al. Compulsive buying: descriptive characteristics and psychiatric comorbidity. *J Clin Psychiatry* 1994; 55: 5–11.
 58. Zuzárrregui JRP and During EH. Sleep issues in Parkinson's disease and their management. *Neurotherapeutics* 2020; 17: 1480–1494.
 59. Zhu K, Van Hilten JJ and Marinus J. The course of insomnia in Parkinson's disease. *Parkinsonism Relat Disord* 2016; 33: 51–57.
 60. Trenkwalder C, Kohnen R, Högl B, et al. Parkinson's disease sleep scale—validation of the revised version PDSS-2. *Mov Disord* 2011; 26: 644–652.
 61. Kurtis MM, Balestrino R, Rodriguez-Blazquez C, et al. A review of scales to evaluate sleep disturbances in movement disorders. *Front Neurol* 2018; 9: 369.
 62. Numbeo. Pollution comparison between Egypt and Morocco, https://www.numbeo.com/pollution/compare_countries_result.jsp?country1=Egypt&country2=Morocco (accessed 26 October 2024).
 63. Högl B, Arnulf I, Comella C, et al. Scales to assess sleep impairment in Parkinson's disease: critique and recommendations. *Mov Disord* 2010; 25: 2704–2716.
 64. Suleiman KH, Yates BC, Berger AM, et al. Translating the Pittsburgh Sleep Quality Index into Arabic. *West J Nurs Res* 2010; 32: 250–268.

65. Attal BA, Al-Ammar FK and Bezdan M. Validation of the Arabic version of the Epworth Sleepiness Scale among the Yemeni medical students. *Sleep Disord* 2020; 2020: 6760505.
66. Nomura T, Inoue Y, Kagimura T, et al. Utility of the REM sleep behavior disorder screening questionnaire (RBDSQ) in Parkinson's disease patients. *Sleep Med* 2011; 12: 711–713.
67. Postuma RB, Arnulf I, Hogl B, et al. A single-question screen for REM sleep behavior disorder: a multicenter validation study. *Mov Disord* 2012; 27: 913–916.
68. Shalash AS, Elrassas HH, Monzem MM, et al. Restless legs syndrome in Egyptian medical students using a validated Arabic version of the Restless Legs Syndrome Rating Scale. *Sleep Med* 2015; 16: 1528–1531.
69. Ismail A, Chabbouh A, Charro E, et al. Examining the validity and reliability of the Arabic translated version of the depression and somatic symptoms scale (A-DSSS) among the Lebanese adults. *Sci Rep* 2024; 14: 5435.
70. Brown K and Ogilvie S. Concise encyclopedia of languages of the world. *Elsevier Science* 2010; 54.
71. Kamusella T. The Arabic language: a Latin of modernity? *Journal of Nationalism, Memory & Language Politics* 2017; 11: 117–145.
72. Chaudhuri KR, Schrag A, Weintraub D, et al. The Movement Disorder Society Nonmotor Rating Scale: initial validation study. *Mov Disord* 2020; 35: 116–133.
73. Gallagher DA, Goetz CG, Stebbins G, et al. Validation of the MDS-UPDRS Part I for nonmotor symptoms in Parkinson's disease. *Mov Disord* 2012; 27: 79–83.
74. Marttila RJ and Rinne UK. Progression and survival in Parkinson's disease. *Acta Neurol Scand Suppl* 1991; 136: 24–28.
75. Levis B, Benedetti A, Ioannidis JPA, et al. Patient Health Questionnaire-9 scores do not accurately estimate depression prevalence: individual participant data meta-analysis. *J Clin Epidemiol* 2020; 122: 115–128.e1.